
NOTES ON A DEGENERATE PARABOLIC EQUATION

Based on *Uniqueness and Root-Lipschitz
Regularity for a Degenerate Heat Equation*

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1 Introduction

Definition 1.1 (Heat Equation and Adjoint equation). The heat equation

$$\partial_t u = \Delta u, \quad u \in \mathbb{R}^n \times \mathbb{R}$$

has a class of solution

$$\Gamma(x, t; \xi, \tau) = \frac{1}{(2\sqrt{\pi})^n} (t - \tau)^{-n/2} \exp\left(-\frac{|x - \xi|^2}{4(t - \tau)}\right), \quad (\xi, \tau) \in \mathbb{R}^n \times \mathbb{R}$$

where for any continuous function $f(x) \lesssim \exp(h|x|^2)$ for some $h > 0$, then the integral

$$u(x, t) = \int \Gamma(x, t; \xi, 0) f(\xi) d\xi$$

exists for $0 < t \leq T$ if $T < 1/4h$ and $u(x, t) \rightarrow f(x), t \rightarrow 0$, and Γ is called a fundamental solution of the heat equation; and u is a solution of the Cauchy problem, i.e. the problem of finding a solution to the heat equation for $0 < t \leq T$ with the initial condition $u(x, 0) = f(x)$.

The adjoint equation is defined as

$$\partial_t v = -\Delta v, \quad v \in \mathbb{R}^n \times \mathbb{R}$$